Research Notes on Flow-based generative model

Flow-based generative model

>> Section 1

R gcal Δ (p ; A , c) = | det ■ (A) | | λ | ∏ i = 1 n q i p i {\displaystyle R_{\text{gcal}}^{\Delta }(\mathbf {p} ;\mathbf {A} ,\mathbf {c})={\frac {\left|\operatorname {det} (\mathbf {A})\right|}{|\lambda |}}\prod _{i=1}^{n}{\frac {q_{i}}{p_{i}}}}

>> Section 2

By a laborious manipulation of R f (x) = | det ■ (T y ′ F x T x) | {\displaystyle R_{f}(\mathbf {x})=\left|\operatorname {det} (\mathbf {T_{y}} '\mathbf {F_{x}T_{x}})\right|} . By setting M = I n {\displaystyle \mathbf {M} =\mathbf {I} _{n}} in R aff (x ; M , c) {\displaystyle R_{\text{aff}}(\mathbf {x} ;\mathbf {M} ,\mathbf {c})} , which is given below. Finally, it is worth noting that f trans {\displaystyle f_{\text{trans}}} and f trans − 1 {\displaystyle f_{\text{trans}}^{-1}} do not have the same functional form.

>> Section 3

The positive orthant of R n {\displaystyle \mathbb {R} ^{n}} is projected onto the simplex as: π (z) = (∑ i = 1 n z i) − 1 z {\displaystyle \pi (\mathbf {z})={\bigl (}\sum _{i=1}^{n}z_{i}{\bigr)}^{-1}\mathbf {z} }

>> Section 4

If f gcal {\displaystyle f_{\text{gcal}}} is to be used as a calibration transform, further constraint could be imposed, for example that A {\displaystyle \mathbf {A} } be positive definite, so that (A x) ′ x > 0 {\displaystyle (\mathbf {Ax})'\mathbf {x} >0} , which avoids direction reversals. (This is one possible generalization of a > 0 {\displaystyle a>0} in the f cal {\displaystyle f_{\text{cal}}} parameter.) For n = 2 {\displaystyle n=2} , a > 0 {\displaystyle a>0} and A {\displaystyle \mathbf {A} } positive definite, then f cal {\displaystyle f_{\text{cal}}} and f gcal {\displaystyle f_{\text{gcal}}} are equivalent in the sense that in both cases, log ■ p 1 p 2 ■ log ■ q 1 q 2 {\displaystyle \log {\frac {p_{1}}{p_{2}}}\mapsto \log {\frac {q_{1}}{q_{2}}}} is a straight line, the (positive) slope and offset of which are functions of the transform parameters. For n > 2 , {\displaystyle n>2,} f gcal {\displaystyle f_{\text{gcal}}} does generalize f cal {\displaystyle f_{\text{cal}}} . It must however be noted that chaining multiple f gcal {\displaystyle f_{\text{gcal}}} flow transformations does not give a further generalization, because:

>> Section 5

Downsides Despite normalizing flows success in estimating high-dimensional densities, some downsides still exist in their designs. First of all, their latent space where input data is projected onto is not a lower-dimensional space and therefore, flow-based models do not allow for compression of data by default and require a lot of computation. However, it is still possible to perform image compression with them. Flow-based models are also notorious for failing in estimating the likelihood of out-of-distribution samples (i.e.: samples that were not drawn from the same distribution as the training set). Some hypotheses were formulated to explain this phenomenon, among which the typical set hypothesis, estimation issues when training models, or fundamental issues due to the entropy of the data distributions. One of the most interesting properties of normalizing flows is the invertibility of their learned bijective map. This property is given by constraints in the design of the models (cf.: RealNVP, Glow) which guarantee theoretical invertibility. The integrity of the inverse is important in order to ensure the applicability of the change-of-variable theorem, the computation of the Jacobian of the map as well as sampling with the model. However, in practice this invertibility is violated and the inverse map explodes because of numerical imprecision.

>> Section 6

Since the trace depends only on the diagonal of the Jacobian ∂ z t f {\displaystyle \partial _{z_{t}}f} , this allows "free-form" Jacobian. Here, "free-form" means that there is no restriction on the Jacobian's form. It is contrasted with previous discrete models of normalizing flow, where the Jacobian is carefully designed to be only upper- or lower-diagonal, so that the Jacobian can be evaluated efficiently.

>> Section 7

e x (x ~) = π (x + T x x ~) , with Jacobian: E x = T x at x ~ = 0 . {\displaystyle e_{\mathbf {x} }({\tilde {x}})=\pi (\mathbf {x} +\mathbf {T_{x}}{\tilde {x}}})\,,{\text{with Jacobian:}}\,\mathbf {E_{x}} =\mathbf {T_{x}} \,{\text{at}}\,{\tilde {\mathbf {x} }}=\mathbf {0} .}

>> Section 8

R aff (x ; M , c) = R lin (x ; M + c x ′) = | det ■ M | (1 + x ′ M − 1 c) ■ M x + c ■ n {\displaystyle R_{\text{aff}}(\mathbf {x} ;\mathbf {M} ,\mathbf {c})=R_{\text{lin}}(\mathbf {x} ;\mathbf {M} +\mathbf {c} \mathbf {x} ')={\frac {\left|\operatorname {det} \mathbf {M} \right|(1+\mathbf {x} '\mathbf {M^{-1}c})}{\lVert \mathbf {Mx+c} \rVert ^{n}}}}

>> Section 9

e : p ~ = (p 1 … , p n − 1) ■ p = (p 1 … , p n − 1 , 1 − ∑ i = 1 n − 1 p i) {\displaystyle e:{\tilde {\mathbf {p} }}=(p_{1}\dots ,p_{n-1})\mapsto \mathbf {p} =(p_{1}\dots ,p_{n-1},1-\sum _{i=1}^{n-1}p_{i})}

>> Section 10

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Generalized calibration transform The above calibration transform can be generalized to f gcal : Δ n − 1 → Δ n − 1 {\displaystyle f_{\text{gcal}}:\Delta ^{n-1}\to \Delta ^{n-1}} , with parameters c ∈ R n {\displaystyle \mathbf {c} \in \mathbb {R} ^{n}} and A {\displaystyle \mathbf {A} } n -by- n {\displaystyle n{\text{-by-}}n} invertible:

>> Section 11

In full Euclidean space, f lin : R n → R n {\displaystyle f_{\text{lin}}:\mathbb {R} ^{n}\to \mathbb {R} ^{n}} is not invertible, but if we restrict the domain and co-domain to the unit sphere, then f lin : S n − 1 → S n − 1 {\displaystyle f_{\text{lin}}:\mathbb {S} ^{n-1}\to \mathbb {S} ^{n-1}} is invertible (more specifically it is a bijection and a homeomorphism and a diffeomorphism), with inverse f lin (· ; M − 1) {\displaystyle f_{\text{lin}}(\cdot \,;\mathbf {M} ^{-1})} . The Jacobian of f lin : R n → R n {\displaystyle f_{\text{lin}}:\mathbb {R} ^{n}\to \mathbb {R} ^{n}} , at y = f lin (x ; M) {\displaystyle \mathbf {y} =f_{\text{lin}}(\mathbf {x} ;\mathbf {M})} is ■ M x ■ − 1 (I n − y y ′) M {\displaystyle \lVert \mathbf {Mx} \rVert ^{-1}(\mathbf {I} _{n}-\mathbf {yy} ')\mathbf {M} } , which has rank n − 1 {\displaystyle n-1} and determinant of zero; while as explained here, the factor (see subsection below) relating source and transformed densities is: ■ M x ■ − n | det ■ M | {\displaystyle \lVert \mathbf {Mx} \rVert ^{-n}\left|\operatorname {det} \mathbf {M} \right|} .